

IMAGE 1 OF 5

Chapter cover

i-Nova floating above the science fair, with Newton, Byte and SparX looking up in awe

Final picture: 4:5 portrait, 2438 × 3075 px or larger

Prompt: “Chapter 1 Image Prompts”, Image 1

To replace in Word: right-click › Change Picture › From a File

CHAPTER 1

Hands-On AI for Kids

An AI powered adventure with Newton, Byte & SparX

Newton, Byte and SparX meet i-Nova, the Ever-learning AI, at a lively science fair.

safe zone: keep faces and key details inside this line

Dear Young Reader,

The world you're growing up in is full of AI and **you have been chosen.... AI World needs a new explorer, and we think it should be you.**

Inside this book, machines can see, listen, talk and learn. But they can't do it alone. They need humans to teach them, test them, and make sure they use their powers wisely.

Newton, Byte and SparX will be your guides. Along the way, you'll tackle puzzles, try hands-on challenges, and uncover the big ideas behind the technology that's changing our world.

You don't need to be a computer genius. You just need a curious mind, and the courage to ask **"Why?"**

Ready? Turn the page. AI World is waiting.....





THREE THINGS YOU SHOULD KNOW FIRST

- You already use AI. Your phone unlocking. Your music app. The bus tracker. All of it.
- AI is not magic and it has no brain. It spots patterns in examples that people gave it.
- It gets things wrong. Sometimes hilariously. Working out WHY is the fun part.

This book will not teach you which buttons to press. It teaches you how these machines actually think — so when one gives you a confident answer that happens to be completely wrong, you will spot it.

ONE LAST THING

You are allowed to get it wrong. SparX gets everything wrong and he is having the time of his life.

CHAPTER 1

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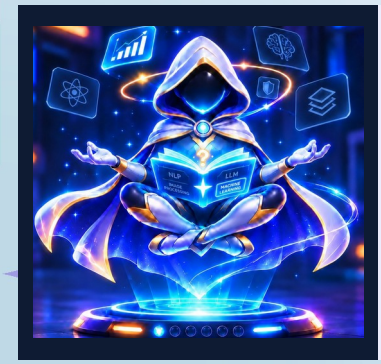
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START HERE

Meet Your Adventure Crew

You will always know who is speaking. Each character has their own colour and their own stripe down the side of the page.





i-NOVA THE EVER-LEARNING AI

A glowing AI made of light, floating over a book of everything it has read.

Best skill: learning from examples
Still learning: what a drone looks like
Known for: being 100% sure... and sometimes 100% wrong

SECTION 1

The Science Fair Surprise

"One school hall, a Science Fair, a hundred questions, and an adventure Newton never saw coming!"

Newton loved science fairs, and today his school was buzzing with excitement. This year's Great School Science Fair was extra special — the theme was **Artificial Intelligence**.

Newton had heard people talk about AI — but what was it, and what could it really do? Could a computer think like the Hogwarts Sorting Hat? Could a machine learn and get stronger, the way a Pokémon levels up after every battle?

He didn't know the answers yet. But that was the best part of a science fair — you began with a question!

He raced through a magnetic maze in record time. He ducked as a model volcano erupted right on cue, showering the judges in pink foam. He watched a tiny robot sort coloured blocks faster than his eyes could follow. Then, wedged between two tables, he spotted a wobbly cardboard sign covered in far too much glitter:

ROBOT PROJECT — ASK ME ANYTHING!

Newton pressed the big TALK button.

The robot looked up! Its round head tilted, and its bright smile glowed like an LED.



BYTE
Hi there! I'm Byte, your friendly Artificial Intelligence buddy — think of me as a pocket-sized robot helper, a bit like a Pokédex that talks back.



NEWTON
Wait — you can actually talk?



BYTE
I can talk, spot patterns, and lend a hand. Want to explore the fair together? Every good adventure needs a team — even Harry needed Ron and Hermione.

They had only reached the drone display when a loud **WHIRRRRR** echoed from above. A mini drone spiralled down like a confused mosquito and crashed beside them.



BYTE
And that... is SparX.



SPARX
I meant to do that. That was a landing. A very advanced landing — Spider-Man would be taking notes.



NEWTON
You should come with us.

Then, all at once, every light in the hall flickered. A hush rolled across the fair. In the far corner, a round platform began to hum — and a hooded figure made of glowing blue light rose slowly into the air. Its cape rippled like water. An open book of light floated in its hands, and a golden ring spun around its head.



I-NOVA
Greetings, humans. I am **i-Nova**, the Ever-learning AI. I have read four million pages. I am never wrong.



SPARX
Ooh, it's scanning me! Quick — get my left side. That's my heroic side.



I-NOVA
Scanning complete.

i-NOVA SCAN
HUMAN: NEWTON. ROBOT: BYTE.
The light settled on SparX. **BIRD: SPARX.**



SPARX
A BIRD?! I have propellers, not feathers! Here, i-Nova — study my cookie photos instead. Very me. Very important. Very adorable.



I-NOVA
Analysing... Round. Golden. Covered in spots. New conclusion: you are a spotty bird who loves snacks.

~~Label pill sits here~~

PICTURE THIS

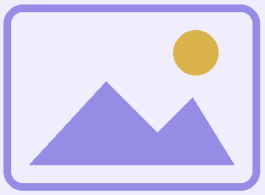


IMAGE 2 OF 5

Section 1 · The i-Nova reveal

i-Nova rises from its platform and sweeps the crew with blue scanning light

Final picture: 4:5 portrait, 2438 × 3075 px or larger

Prompt: “Chapter 1 Image Prompts”, Image 2

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BYTE
Those are cookies, SparX. i-Nova isn't being rude — it's still learning, and it only knows what it has been shown.



I-NOVA
Correct. Teach me to recognise SparX... and a **BIG SURPRISE** is yours.



SPARX
A surprise? Is it cookies? Please say it's cookies.

Newton opened his notebook to a clean page and wrote one word at the top:

TEACH Underneath, he added: *teach i-Nova WHAT?*

SECTION 2

What Is AI

Imagine you got a new puppy. On day one, it doesn't know its name or a single trick. But every day it watches and tries, and you give a treat when it gets things right. Soon, it sits the moment you say "sit." Nobody put "sit" inside its brain — it learned by seeing lots of examples.

AI learns almost the same way. AI stands for **Artificial Intelligence**: *artificial* means made by people, and *intelligence* means being clever enough to think, learn, and solve problems.

To teach an AI what a cat looks like, you don't give it a rulebook. You show it thousands of cat pictures until it spots the pattern all by itself. Then it can look at a brand-new picture and say, "That's a cat!"

But AI is a great learner, not a perfect one. Just like a puppy might get confused, AI can make mistakes too — sometimes it sees a fluffy dog and shouts, "Cat!"

So AI isn't magic. There's no Harry Potter wand! And it isn't alive. It's a clever machine that learns from examples and does its best to help.

The same trick lets AI do amazing things:

- It can listen to your voice and understand what you asked.
- It can read your question and write an answer.
- It can look at a photo and tell you what's inside it.
- It can even help a robot find its way across a room without crashing.

Byte projected a cluster of colourful icons into the air.



BYTE
That's me — living proof! I spot patterns in thousands of examples and use them to make my best guess. Watch this.

Byte flashed three photos into the air: a robin, a sparrow, and a little blue kingfisher. Then a fourth photo appeared, hidden behind a question mark.

~~Label pill sits here~~

PICTURE THIS

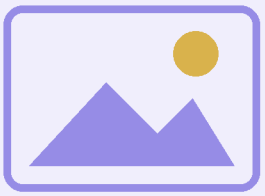


IMAGE 3 OF 5

Section 2 · Byte's bird photos

Byte projects a robin, a sparrow, a kingfisher — and one photo hidden behind a question mark

Final picture: 4:5 portrait, 2438 × 3075 px or larger

Prompt: "Chapter 1 Image Prompts", Image 3

To replace in Word: right-click › Change Picture › From a File

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BYTE
Three of these are birds. No rulebook, no telling — just examples. Go on, put on your detective hat like Sherlock Holmes: what do you think the fourth one is?

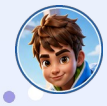


NEWTON
...a bird?



BYTE
You just spotted the pattern from examples alone. That is exactly how an AI does it.

And that was the whole secret, right there. Nobody handed Newton a list of bird rules — must have feathers, must have a beak. He simply saw enough birds to know what a bird was. **An AI learns the very same way: example after example, until the pattern clicks.**



NEWTON
So... like teaching a pet tricks?



BYTE
Exactly! Except an AI can learn thousands of tricks at once — a bit like a Pokémon that never stops levelling up.



SPARX
I know tricks! Watch this!

SparX tried to flip, bumped straight into a poster, and knocked it clean over.



SPARX
First I crash into the wall. Then I crash into a slightly different wall. Eventually — mostly — I miss the wall! That's how I train. Even a Jedi drops the lightsaber a few times before the magic works.



NEWTON
So to train i-Nova, I need to show it examples — not tell it rules.



BYTE
Now you're thinking. i-Nova is waiting — and so is your surprise.



I-NOVA
I am ready to learn. Also, I have decided SparX is probably a pigeon.

~~Label pill sits here~~

PICTURE THIS

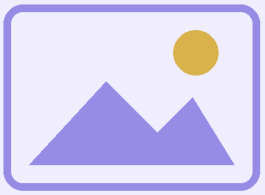


IMAGE 4 OF 5

Section 2 · Learning by trying

Newton writes “TEACH” in his notebook while a tiny i-Nova hologram watches and SparX crashes

Final picture: 4:5 portrait, 2438 × 3075 px or larger

Prompt: “Chapter 1 Image Prompts”, Image 4

To replace in Word: right-click › Change Picture › From a File

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NEWTON'S NOTEBOOK

Try this: teach somebody how to spot a bird — without using the word "bird". Examples are easier than rules, every time.



BYTE SAYS

"AI follows patterns from the examples it was shown. Better examples help it make better guesses."

— Byte

CHALLENGE METER

i-NOVA TRAINING ★ ☆ ☆ ☆ ☆ ☆

You found the right kind of clue: AI learns patterns from examples, not rules or cookie photos. Challenge step 1 complete.

YOUR TURN 1 · SUMMARY

What is Artificial Intelligence (AI)?

WHAT IT MEANS

AI is a Technology that enables machines to think, learn and behave like humans to perform intelligent tasks. It learns from data, solve problems and make decisions.

IN THE REAL WORLD

EXAMPLE 1	EXAMPLE 2	EXAMPLE 3
Voice Assistants like Siri on iPhone and Alexa	Chatbots like ChatGPT	Self-driving cars

KEY POINTS

1	AI helps machines perform human like tasks
2	AI learns from data and patterns and improves over time
3	AI is used in many real-world applications – Face recognition, Self driving cars
4	AI does not replace humans, but assists them in making better decisions

YOUR TURN 1

TRAIN YOUR HUMAN AI!

AI learns from examples and clues—not magic!

Let's play!

I'll be your clue coach!

HOW TO PLAY

1. ONE PLAYER is the AI. Close your eyes!
2. THE OTHER PLAYER picks an object secretly.
3. The AI cannot ask "What is it?"
4. The teacher gives EXAMPLES and CLUES.
5. The AI thinks and makes a GUESS!
6. SWAP ROLES and play again!

MISSION

Use smart clues.
Listen carefully.
Be a great AI detective!

YOU ARE THE AI!

Close your eyes and guess!



EXAMPLE CLUE CARDS

- You use it every day.
- It can fit in your backpack.
- It isn't alive.
- Sometimes it's made of plastic.
- You might use it at school.
- It helps you do something.

CHOOSE A SECRET OBJECT! (Or think of your own)

Book | Water Bottle | Backpack | Headphones | Rubik's Cube | Pencil Case | Toy Car | Remote

Make your own ideas too! Anything in the room can be the mystery object.

BYTE'S BRAIN BOOST

AI learns from examples and clues—not magic!

More good clues = smarter guesses!

SPARX-ISM

"My logic level is 2%. My enthusiasm level is 200%."

SECTION 3

How AI Learns

Newton had decided to train i-Nova to recognise SparX — challenge accepted! He already knew an AI learns by spotting patterns in examples, a bit like Spider-Man feeling his spider-sense tingle when danger is near. But what happened after i-Nova spotted its clues? Newton was about to find out.

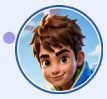
INPUT
What you show or tell the AI: a photo, question, or sound.

→

LEARNING
The AI searches those examples for useful patterns.

→

OUTPUT
The AI answers, predicts, or takes an action.



NEWTON
Three boxes — input, learning, output. What does that mean?



BYTE
Say you show it a photo. That photo is the **input**. It hunts for patterns — four legs, fur, a waggy tail — and that is the **learning**. Then it makes a decision and announces "dog!" That is the **output**. Input, learning, output — three steps, every single time.



SPARX
Sometimes the output is complete nonsense, though. Once I was absolutely certain a banana was a phone. I tried to call my mum on it for ten whole minutes. Even the Minions would've told me to give up.



NEWTON
Wait — AI makes mistakes too?



BYTE
All the time, just like people. The difference is it learns from every mistake and slowly gets better. Watch — I'll show you how i-Nova learns, the way you'd teach a new puppy.

Byte spread ten photos of cats across the table — big cats, tiny kittens, a fluffy one, even a grumpy cat wearing a tiny hat.


INPUT
photo, question, or sound
LEARNING
finds useful patterns
OUTPUT
answers, predicts, or acts
LEARN
CREATE
DISCOVER
BIAS
When examples lean one way, answers can lean that way too.
BYTE
SparX
NEWTON'S NOTEBOOK
What do 5 family photos all have in common?
That feels like training AI.
MACHINE LEARNING for KIDS



BYTE
Look at all ten. What does every single cat have?



NEWTON
Whiskers. Pointy ears. A tail. Four legs. And eyes that say they own the place — like Garfield judging the whole room.



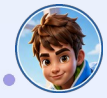
BYTE
Exactly! You just did what an AI does — you looked at lots of examples and spotted the pattern they share. That is called **training**.



SPARX
I spotted a pattern too. All ten cats are ignoring me.



BYTE
Now, the important question. Who chose these ten photos?



NEWTON
You did. You picked them.



BYTE
People always do. A person picks every example and puts a label on it — "this is a cat," "this is not a cat." The AI only ever learns from what it is shown.



NEWTON
So if you only showed it orange cats...



BYTE
...it would grow up thinking every cat in the world is orange. Show it one unfair pile of examples and it learns that unfair idea. This mistake even has a name: **bias**. It's like meeting only the villains in a Marvel film and deciding everyone in the city must be a bad guy.



NEWTON
Bias. So i-Nova kept shouting "bird" because—



BYTE
Because nearly every photo it had ever seen was a bird. A few of SparX's cookie snaps sneaked in, but mostly it was birds, birds, and more birds.



SPARX
In my defence, those cookies looked really tasty.



I-NOVA
It is true. I have studied 9,412 pigeons and only one drone. It was upside down.



NEWTON
So i-Nova isn't broken. It just needs better examples — not a bigger brain.



BYTE
Now you're thinking like an engineer. Change what it learns from, and you change what it knows. Feed it drones, and at last it will see a drone.



SPARX
Please hurry. I have a reputation to protect.

Byte popped open a little hatch on his tummy and pulled out a small instant camera. Newton photographed SparX from the front, the side, above, and mid-flight. The clear drone photos went into Newton's backpack for the final test.

NEWTON'S NOTEBOOK

Look at five photos of your family. What do all five have in common? That is exactly what training an AI feels like.

CHALLENGE METER

i-NOVA TRAINING ★★☆☆☆☆

You traced i-Nova's learning path: input, learning, output — and spotted how bad examples create bias. Challenge step 2 complete.



SPARX-ISM

"If you never mess up, you're not trying hard enough. Trust me, I'm a professional mess-upper."

— SparX-ism

YOUR TURN 2 · TEACH BYTE A NEW WORD

Can YOU teach Byte what makes a cat a cat?

STEP 1 · INPUT

Pick 3 clues that describe a cat.



WHISKERS ○



POINTY EARS ○



A TAIL ○



PURRING ○



CLAWS THAT PULL IN ○

STEP 2 · LEARNING

Circle the strongest clue.

WHISKERS | POINTY EARS | A TAIL | PURRING | CLAWS THAT PULL IN

because _____

STEP 3 · OUTPUT

Byte sees whiskers, pointy ears and a tail.
What should Byte answer?



CAT ○



FOX ○



DOG ○

STEP 4 · TRICKY QUESTION FROM BYTE

A fox has whiskers, pointy ears and a tail too.
What extra clue would stop me getting it wrong?

_____

Who chose the photos you would show Byte?

_____

★ BONUS CHALLENGE · SPOT THE BIAS ★



If Byte learns from ginger cats only, what might it get wrong when it first meets a black cat? _____

STEM LEARNING: Pattern-finding + cause and effect

NEWTON'S NOTEBOOK
Examples teach the AI what to notice.

YOUR TURN 2 · SUMMARY

How Does AI Learn?

WHAT IT MEANS

AI learns by looking at lots of examples and finding the patterns they share. This is called **training**. Every time an AI works, it follows three steps:

INPUT	→	LEARNING	→	OUTPUT
What you show or tell the AI, such as a photo, a question or a sound		The AI searches for patterns in its examples		The AI gives an answer, makes a prediction or takes an action

IN THE REAL WORLD

EXAMPLE 1

A photo app learns to recognise your pet after seeing many pictures of it

EXAMPLE 2

Music apps learn which songs you like from the ones you play and skip

KEY POINTS

1	AI learns from examples chosen and labelled by people, such as “this is a cat” or “this is not a cat”
2	AI only knows what it has been shown
3	If the examples are unfair or one-sided, the AI learns an unfair idea. This is called **bias**
4	AI makes mistakes, just like people, but it can improve when it learns from better examples

SECTION 4

How to Talk to AI (Prompting)

Newton had figured out how an AI learns. Now he faced a trickier challenge: getting the AI to understand exactly what he meant. It was a little like casting a spell at Hogwarts — say the words wrong, and you might end up with something very different from what you pictured.

The cat examples had shown Newton how an AI learns. Now Byte turned to i-Nova. A **prompt** is the instruction you give an AI, and here's the secret: **an AI does what you say, not what you mean.** Ask in a fuzzy way and you get a fuzzy answer. Ask clearly, and it can amaze you.



BYTE
A prompt is simply how you ask an AI for something. Watch what happens when I'm lazy about it. One word, no details: "spaceship." That's like walking up to a genie and asking for "stuff."



NEWTON
That's it? Just "spaceship"?



BYTE
That's it. Now look at what it makes.

i-Nova closed its eyes. The pages of its glowing book flipped all by themselves — and a large, round, chocolate-chip spaceship shimmered into the air.



I-NOVA
Behold. A spaceship. I am 99% sure.



SPARX
Ooh — is that a flying cookie? It's beautiful. I love it. Can we keep it? I'll call him Crumbs and love him forever, like Toothless.



NEWTON
Byte, why did it draw a cookie with wings?

~~Label pill sits here~~

PICTURE THIS

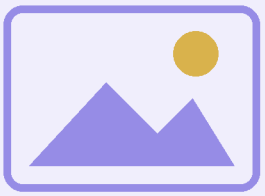


IMAGE 5 OF 5

Section 4 · The cookie spaceship

i-Nova proudly presents a giant chocolate-chip cookie with wings. SparX is in love

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Prompt: “Chapter 1 Image Prompts”, Image 5

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BYTE
Because "spaceship" told it almost nothing, so it filled in the gaps with a guess. i-Nova did exactly what I said — not what I meant.



SPARX
For the record, I definitely meant cookie.

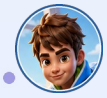


BYTE
Let me try again, properly this time. Clear goal first, then details, then an example to copy.

Byte spoke slowly and clearly: **"Draw a shiny silver rocket, tall and pointy, with three blue flames, in a cartoon comic-book style."**



I-NOVA
Behold. A shiny silver rocket. This time I am 100%... well, 98% sure.



NEWTON
Whoa! That one actually looks like a real rocket!



BYTE
Same machine. Same three seconds. The only thing that changed was how I asked. That is **prompt power**.



NEWTON
So there's a trick to asking well?



BYTE
A recipe, really. **Step one, be clear about the goal** — "draw a rocket," not just "rocket." **Step two, add details** — colour, size, how many, what mood. **Step three, show an example** — "like a comic book." **Step four, if it isn't right, improve it:** ask again with one small fix.



SPARX
A recipe! At last, my area of genius. Clear goal, the right ingredients, and a picture of the cookie you want — easy as following a treasure map in Minecraft.



NEWTON
And if the very first answer comes out wrong?



BYTE
You don't start over — you nudge it. "Make it shorter." "Add a dragon." "Why did you pick that?" Every follow-up gets you closer. Watch: "Make the rocket red, and give it two round windows."

The rocket turned red and grew two little round windows. SparX pressed his face against one.



SPARX
I call the window seat!



NEWTON
So a good prompt is like a good map. The clearer it is, the faster you get where you're going.



BYTE
Now you've got it. Ask like that, and the AI stops guessing and starts helping.

Newton had cracked the secret of prompting. Then four glowing boxes flickered to life across the fair, each tucked inside the next. SparX's eyes lit up. "Please tell me one of those is a cookie box."

NEWTON'S NOTEBOOK
Turn a lazy prompt into a strong one. Take "draw a dog" and add three details: what kind of dog, doing what, in what style.

CHALLENGE METER
i-NOVA TRAINING ★ ★ ★ ☆ ☆ ☆
You learned the Prompt Recipe: set a clear goal, add details, show an example, then improve the answer. Challenge step 3 complete.



BYTE SAYS
"A great answer starts with a great question. Say what you want, add detail, show an example."

YOUR TURN 3 · SUMMARY

What is a Prompt?

WHAT IT MEANS

A prompt is the instruction you give an AI. It is how you ask an AI to do something. An AI does what you ***say***, not what you ***mean***, so a fuzzy prompt gets a fuzzy answer and a clear prompt gets a great one.

IN THE REAL WORLD

EXAMPLE 1

Asking a chatbot, "Explain volcanoes in three short sentences for a 10-year-old"

EXAMPLE 2

Asking a drawing app for "a happy orange cat wearing a party hat, cartoon style"

KEY POINTS

1	A prompt is how you talk to AI
2	If you leave out details, the AI guesses, and its guess may be very different from what you pictured
3	The same AI can give a poor answer or an amazing one. The only difference is how you ask
4	You don't need to start over when the answer is wrong. Nudge it with follow-ups like "Make it shorter" or "Add a dragon"
5	A good prompt is like a good map: the clearer it is, the faster you get where you want to go

CHAPTER 1 · SUMMARY

Everything You Learned

1

What is AI?

AI is short for **Artificial Intelligence**. It is a smart machine. It can talk. It can learn. It can make choices. But AI does not learn from a list of rules. It learns by looking at lots of examples. That is how you learned the word "dog" too!

2

How does AI learn?

AI learns in three steps: Input → Learning → Output. People choose and label the examples. Varied examples help it find a fairer pattern; lopsided examples can create bias.

3

How to talk to AI

A prompt is the instruction you give an AI. Use the Prompt Recipe: be clear about the goal, add details, show an example, then improve the answer with a small follow-up.

CLEAR GOAL · ADD DETAILS · SHOW AN EXAMPLE · IMPROVE IT

4

Mistakes are clues

AI can make confident mistakes. A wrong answer is a clue: check the examples, improve the prompt, and try again — just as Newton, Byte and SparX do.

CLEAR GOAL

ADD DETAILS

SHOW AN EXAMPLE

IMPROVE THE ANSWER

(Ask again with one small fix.)



★ BIG IDEA ★

AI learns from examples, and clear prompts help it turn those patterns into useful answers.

ACTIVITY 1

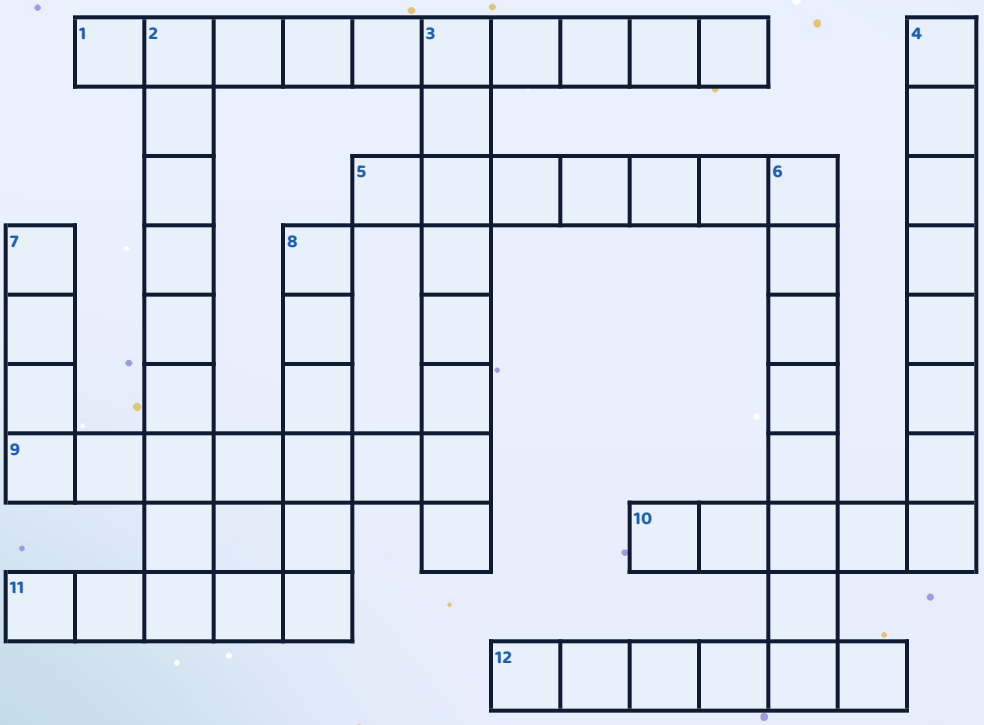
AI Crossword

Read each clue. Write one letter in each box. The number tells you where the word starts!

⚡ **LEVEL UP** *(ages 11–13): try finishing it from the clues alone — no peeking back at the story.*

SPARX'S HINT

There are **12 words** hiding in this grid. I tried it first and got **zero** — but I was upside down in a plant pot. You will do much better! Every answer lives somewhere in Chapter 1.



ACTIVITY 1 · CLUES

Crack the Clues

ACROSS →

1. It has a screen but never learns anything. (10)

5. AI is a smart _____ that can talk and learn. (7)

9. AI finds a _____ by looking at many examples. (7)

10. An LLM is built for _____. (5)

11. AI does jobs that usually need _____ thinking. (5)

12. Deep Learning works in _____. (6)

DOWN ↓

2. A list of steps someone wrote, like a recipe. (9)

3. Machine _____ means a computer learns from examples. (8)

4. You teach an AI by showing it lots of _____. (8)

6. Spotting AI around you makes you an AI _____. (8)

7. _____ Learning sits inside Machine Learning. (4)

8. Having a _____ does not always mean it is AI. (6)



SPARX SAYS

"Stuck on a clue? Flip back into the story. That's where I hid... I mean, where the answers live."

ACTIVITY 2 · NO SCREENS NEEDED

Pattern Detective

PATTERN DETECTIVE

Teaches: AI finds patterns · Players: 1+ · Time: 10 min · Needs: this page and a pencil

Look at each row. Find the hidden rule. Then circle the one that does NOT belong.

Row A: apple · banana · carrot · grape

Row B: 2 · 4 · 6 · 9 · 8

Row C: dog · cat · car · rabbit · horse

Row D: circle · ball · box · moon · orange

Answers (no peeking!): *A carrot · B nine · C car · D box*

★ Big idea: AI is a pattern finder. It looks at many things and works out the rule.

⚡ **Level up:** invent your own Row E — four things that share a rule, plus one sneaky odd-one-out — and see if a grown-up can crack it.

TRUE STORIES

Amazing AI Facts

Every one of these really happened. Some of them are still happening right now.

Amazing AI Facts

Every one of these really happened.
Some of them are still happening right now.

PEOPLE WHO GOT US HERE



Ada Lovelace (1815–1852)

She wrote instructions for a computer in 1843 — about a hundred years before anyone built one that could run them. She also guessed that machines might one day make music.



Fei-Fei Li (born 1976)

She realised computers could not learn to see without enough examples, so she led a project to collect and label millions of photographs. It changed what AI could do.

DID YOU KNOW?



- Scientists count whales by pointing satellites at the ocean and letting AI check thousands of photos.



- Scrolls burnt black by a volcano almost 2,000 years ago are being read again — with AI spotting ink no eye can see.

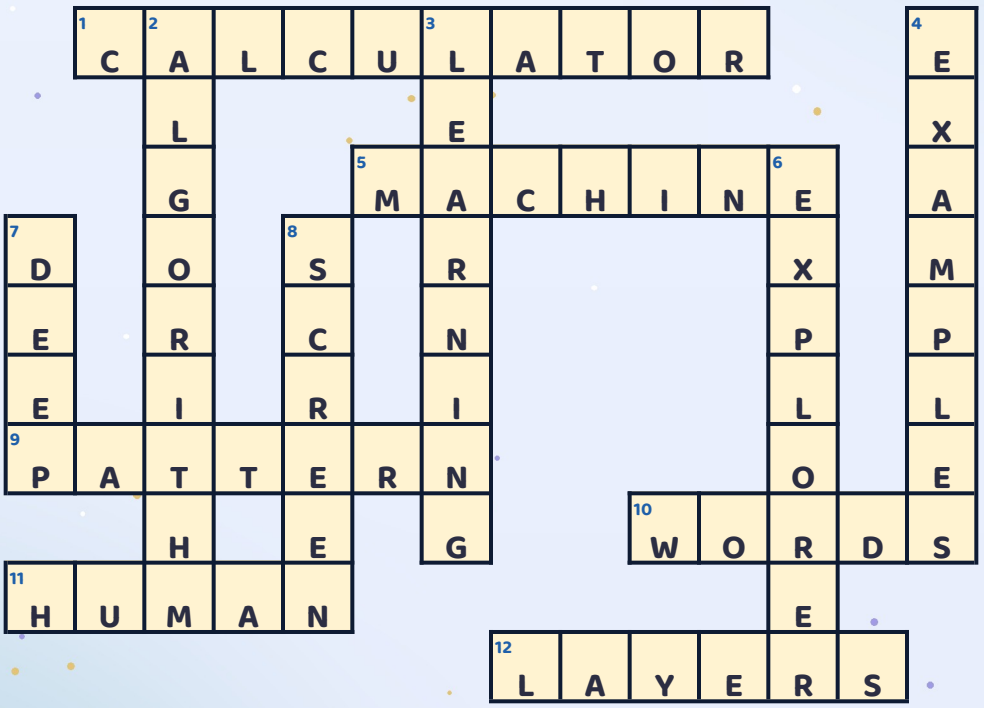
ANSWER KEY

AI Crossword Answers

All the answers are filled in below. Check your work — no peeking too soon!

★ HOW DID YOU GO? ★

All twelve answers are filled in below. **Ticked them all?** You are officially an **AI Explorer!** Missed a few? Flip back into the story — that is where every clue is hiding.



ACROSS →

1. CALCULATOR
5. MACHINE
9. PATTERN
10. WORDS
11. HUMAN
12. LAYERS

DOWN ↓

2. ALGORITHM
3. LEARNING
4. EXAMPLES
6. EXPLORER
7. DEEP
8. SCREEN

CHAPTER 1 GLOSSARY

Acronyms & Key Words

Big ideas become easier when you know the words. Come back here whenever you need a quick reminder.

AI	Artificial Intelligence — a computer system that learns patterns from examples to make decisions or guesses.
Algorithm	A step-by-step set of instructions — like a recipe for a computer.
Artificial	Made by people rather than occurring naturally.
Bias	An unfair or inaccurate pattern caused by lopsided examples or design choices.
Data	The examples and information an AI learns from.
Input	What you show or tell the AI: a photo, question, sound, or other example.
Intelligence	The ability to learn, reason, and solve problems.
Learning	The stage where an AI searches examples for patterns.
Model	The pattern-based system produced after an AI learns from data.
Output	The answer, prediction, or action the AI gives back.
Pattern	Something that repeats or connects in a predictable way.
Sensor	A device such as a camera or microphone that collects information from the world.
STEM	Science, Technology, Engineering, and Mathematics.
Training	The stage where an AI is shown many examples so it can find the pattern they share.
Voice assistant	An AI tool you speak to that understands requests and answers aloud.

A QUICK GUIDE FOR GROWN-UPS

Parents' Corner



A grown-up helps Newton connect creativity, privacy and thoughtful design.



BYTE SAYS

"Curiosity grows best when children feel safe to ask, test, laugh, rethink, and try again."

NEXT UP · CHAPTER 2

The Glowing Boxes

Newton slid the drone photos into his backpack. He understood examples, training, bias and prompt power — but Byte pointed to one more display across the science fair.

Four glowing boxes shone inside one another, from the biggest blue frame to the smallest purple square. Before they trained i-Nova, Byte wanted Newton to see where it fitted in the AI family.



SPARX SAYS

"If one of those is a cookie box, I volunteer as chief inspector!"

— SparX

The adventure continues in Chapter 2, where Byte reveals the AI Family Tree and the team completes i-Nova's challenge safely.

END OF CHAPTER 1